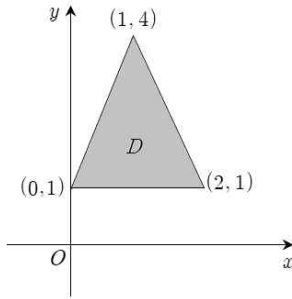


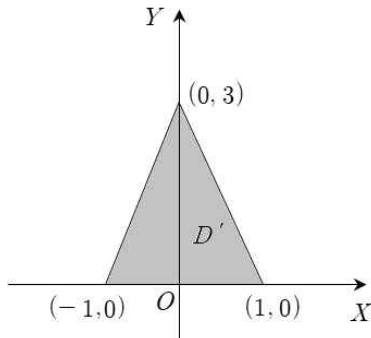
고완풀 단원별 다변수 4단원 해설

【1】

영역을 그려보면 아래와 같다.



$x-1 = X$, $y-1 = Y$ 라 하면 아래와 같이 평행이동한다.

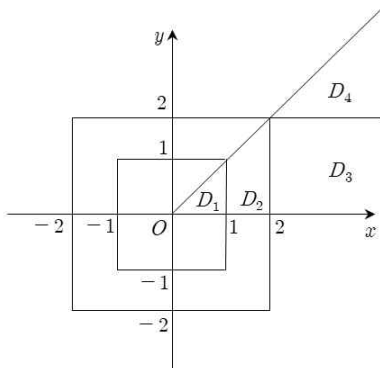


$$\begin{aligned}
 & \iint_D x^2 dx dy \\
 &= \iint_{D'} (X+1)^2 dX dY \\
 &= \iint_{D'} (X^2 + 2X + 1) dX dY \\
 &= \iint_{D'} X^2 dX dY + 2 \iint_{D'} X dX dY + \iint_{D'} 1 dX dY \\
 &= \iint_{D'} X^2 dX dY + D' \text{의 넓이} \quad (\because \iint_{D'} X dX dY = 0) \\
 &= 2 \iint_{D'} X^2 dX dY + 3 \quad (\because D' : D \text{의 1사분면}) \\
 &= 2 \int_0^1 \int_0^{-3X+3} X^2 dY dX + 3 \\
 &= 2 \int_0^1 X^2 (-3X+3) dX + 3 \\
 &= 2 \left(-\frac{3}{4} + 1 \right) + 3 = \frac{7}{2}
 \end{aligned}$$

정답 : ③

【2】

R^2 의 일부 영역을 D_1 , D_2 , D_3 , D_4 라 하자.



$$\begin{aligned}
 & \iint_{R^2} e^{-d(x,y)} dA \\
 &= 8 \left[\iint_{D_1} e^{-d(x,y)} dA + \iint_{D_2} e^{-d(x,y)} dA + \iint_{D_3} e^{-d(x,y)} dA + \iint_{D_4} e^{-d(x,y)} dA \right] \\
 &= 8 \left[\iint_{D_1} e^{-(1-x)} dA + \iint_{D_2} e^0 dA + \iint_{D_3} e^{-(x-2)} dA \right. \\
 & \quad \left. + e^{-\sqrt{(x-1)^2 + (y-1)^2}} dA \right]
 \end{aligned}$$

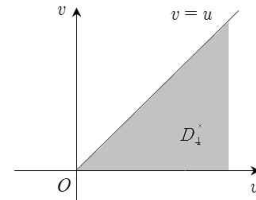
이때,

$$\begin{aligned}
 \iint_{D_1} e^{-(1-x)} dA &= \int_0^1 \int_0^x e^{x-1} dy dx = \int_0^1 x e^{x-1} dx \\
 &= [x e^{x-1} - e^{x-1}]_0^1 = e^{-1},
 \end{aligned}$$

$$\iint_{D_2} 1 dA = D_2 \text{의 넓이} = \frac{1}{2} (1+2) \times 1 = \frac{3}{2},$$

$$\begin{aligned}
 \iint_{D_3} e^{-(x-2)} dA &= \int_2^\infty \int_0^{2-x} e^{2-x} dy dx = \int_2^\infty 2 e^{2-x} dx \\
 &= [-2 e^{2-x}]_2^\infty = 2,
 \end{aligned}$$

$$\begin{aligned}
 & \iint_{D_4} e^{-\sqrt{(x-1)^2 + (y-1)^2}} dA \\
 &= \iint_{D_4'} e^{-\sqrt{u^2 + v^2}} du dv \quad (\because x-1 = u, y-1 = v)
 \end{aligned}$$

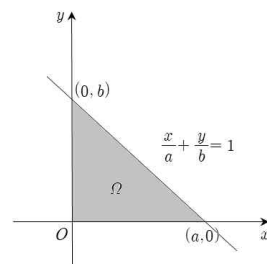


$$= \int_0^{\pi/4} \int_0^\infty e^{-r} r dr d\theta = \int_0^{\pi/4} \frac{1}{2} d\theta = \frac{\pi}{4}$$

$$\text{이므로 } \iint_{R^2} e^{-d(x,y)} dA = 8 \left(e^{-1} + \frac{3}{2} + 2 + \frac{\pi}{4} \right) = 2\pi + 28 + 8e^{-1} \text{이다.}$$

정답 : ⑤

【3】



$$\begin{aligned}
 & \iint_{\Omega} f\left(\frac{x}{a} + \frac{y}{b}\right) dx dy \\
 &= \int_0^a \int_0^{b-\frac{b}{a}x} f\left(\frac{x}{a} + \frac{y}{b}\right) dy dx \\
 &= \int_0^a \int_{\frac{x}{a}}^1 f(u) b du dx \quad (\because \frac{x}{a} + \frac{y}{b} = u) \\
 &= \int_0^1 \int_0^{au} f(u) b dx du \quad (\because \text{적분 순서 변경}) \\
 &= \int_0^1 f(u) abu du = \int_0^1 f(u) g(u) du
 \end{aligned}$$

이므로 $g(u) = abu$ 이다.

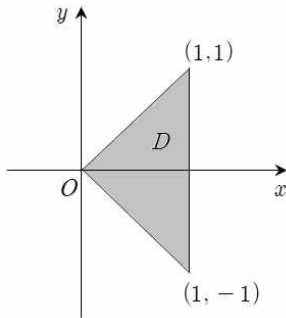
$$\text{즉, } g(ab) = a^2 b^2$$

[다른 풀이]

$u = \frac{x}{a} + \frac{y}{b}$, $v = \frac{x}{a} - \frac{y}{b}$ 라 하면

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{\frac{\partial(u, v)}{\partial(x, y)}} = \frac{1}{\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}} = \frac{1}{\begin{vmatrix} \frac{1}{a} & \frac{1}{b} \\ \frac{1}{a} & -\frac{1}{b} \end{vmatrix}} = \frac{1}{\frac{-2}{ab}} \text{이고,}$$

(x, y) 의 $(0, 0)$, $(a, 0)$, $(0, b)$ 은 (u, v) 의 $(0, 0)$, $(1, 1)$, $(1, -1)$ 으로 대응한다.



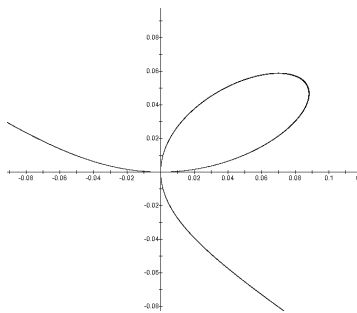
$$\begin{aligned} & \iint_{\Omega} f\left(\frac{x}{a} + \frac{y}{b}\right) dx dy \\ &= \iint_D f(u) \frac{ab}{2} du dv \\ &= \int_0^1 \int_{-u}^u f(u) \frac{ab}{2} dv du \\ &= \int_0^1 f(u) abu du \\ &= \int_0^1 f(u) g(u) du \end{aligned}$$

이므로 $g(u) = abu$ 이다.

즉, $g(ab) = a^2 b^2$

【4】

$8x^3 + 27y^3 = 2xy$ 의 그림을 그려보면 아래와 같고, 1사분면의 둘러싸인 영역을 D 라 하자.



$$\begin{aligned} D \text{의 넓이} &= \iint_D dx dy \\ &= \iint_D \frac{1}{6} du dv \quad (\because 2x = u, 3y = v) \\ &= \frac{1}{6} \times D' \text{의 넓이} \end{aligned}$$

이고, $D' : u^3 + v^3 = \frac{1}{3}uv$ 의 1사분면의 둘러싸인 영역이다.

이때, $x^3 + y^3 = 3axy$ 의 1사분면에 둘러싸인 넓이는 $\frac{3}{2}a^2$ 이므로

D' 의 넓이는 $\frac{3}{2} \times \left(\frac{1}{9}\right)^2$ 이다.

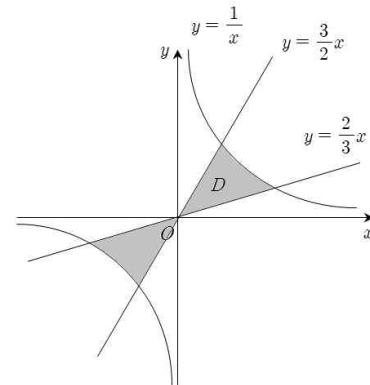
즉, D 의 넓이 $= \frac{1}{6} \times \frac{3}{2} \times \left(\frac{1}{9}\right)^2 = \frac{1}{324}$

정답 : ①

【5】

$$y = \frac{1}{x} \Leftrightarrow xy = 1, \quad y = \frac{3}{2}x \Leftrightarrow \frac{y}{x} = \frac{3}{2},$$

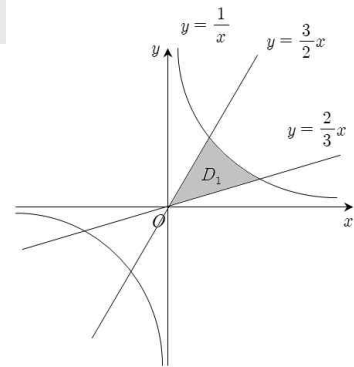
$y = \frac{2}{3}x \Leftrightarrow \frac{y}{x} = \frac{2}{3}$ 의 영역을 그려보면 아래와 같다.



이때, $u = xy$, $v = \frac{y}{x}$ 라 놓으면 일대일 대응이 되지 않는다.

그러므로 D_1 의 영역에 대해서 2배를 적용하자.

$$\text{또한 } J = \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{\frac{\partial(u, v)}{\partial(x, y)}} = \frac{1}{\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}} = \frac{1}{\begin{vmatrix} y & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix}} = \frac{1}{2v} \text{이다.}$$



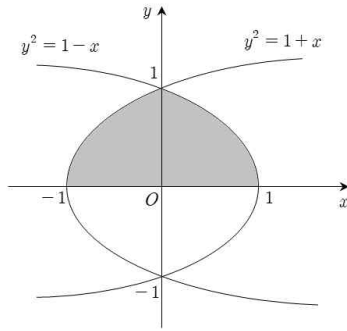
D 의 넓이 $= 2 \times D_1$ 의 넓이

$$\begin{aligned} &= 2 \iint_{D_1} dx dy \\ &= 2 \iint_R \frac{1}{2v} du dv \\ &= \int_{\frac{2}{3}}^{\frac{3}{2}} \int_0^1 \frac{1}{v} du dv \\ &= \int_{\frac{2}{3}}^{\frac{3}{2}} \frac{1}{v} dv = [\ln v]_{\frac{2}{3}}^{\frac{3}{2}} \\ &= \ln \frac{3}{2} - \ln \frac{2}{3} = 2 \ln \frac{3}{2} \end{aligned}$$

정답 : ③

【6】

그림을 그려보면 아래와 같다.



$$\begin{aligned}\iint_R y dA &= \int_0^1 \int_{y^2-1}^{1-y^2} y dx dy \\ &= \int_0^1 y(1-y^2-y^2+1) dy \\ &= \int_0^1 2y-2y^3 dy \\ &= \left[y^2 - \frac{1}{2} y^4 \right]_0^1 = \frac{1}{2}\end{aligned}$$

[다른 풀이]

$x = u^2 - v^2$, $y = uv$ 로 놓으면

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 2u & -2v \\ v & u \end{vmatrix} = 2u^2 + 2v^2 \text{이다.}$$

$$\begin{aligned}\iint_R y dA &= \int_0^1 \int_0^1 uv(2u^2 + 2v^2) du dv \\ &= \int_0^1 \int_0^1 2u^3 v + 2uv^3 du dv \\ &= \int_0^1 \frac{1}{2} v + v^3 dv \\ &= \left[\frac{1}{4} v^2 + \frac{1}{4} v^4 \right]_0^1 = \frac{1}{2}\end{aligned}$$

정답 : ②

【7】

영역 $D : x^2 + y^2 \leq 20$ 은

$$D_1 : 0 \leq x^2 + y^2 < 1,$$

$$D_2 : 1 \leq x^2 + y^2 < 2,$$

$$D_3 : 2 \leq x^2 + y^2 < 3,$$

$\vdots$

$$D_{20} : 19 \leq x^2 + y^2 < 20$$

에 대하여 $D = D_1 \cup D_2 \cup \dots \cup D_{20}$ 이다.

$$\begin{aligned}\iint_D [x^2 + y^2] dA &= \iint_{D_1} [x^2 + y^2] dA + \iint_{D_2} [x^2 + y^2] dA + \dots + \iint_{D_{20}} [x^2 + y^2] dA \\ &= \iint_{D_1} 0 dA + \iint_{D_2} 1 dA + \dots + \iint_{D_{20}} 19 dA \\ &= 0 \times (D_1 \text{ 넓이}) + 1 \times (D_2 \text{ 넓이}) + 2 \times (D_3 \text{ 넓이}) + \dots + 19 \times (D_{20} \text{ 넓이}) \\ &= \pi + 2\pi + 3\pi + \dots + 19\pi \\ &= \frac{19 \times 20}{2} \pi = 190\pi\end{aligned}$$

정답 : ①

【8】

$u = 2x + y$, $v = y$ 라 하면

$$x = \frac{1}{2}(u - v), \quad y = v \text{이고 } J = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 \end{vmatrix} = \frac{1}{2} \text{이다.}$$

$$\begin{aligned}&\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y(x+y) e^{-(4x^2+4xy+2y^2)} dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y(x+y) e^{-[(2x+y)^2+y^2]} dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} v \times \frac{1}{2}(u+v) e^{-(u^2+v^2)} \times \frac{1}{2} du dv \\ &= \frac{1}{4} \int_0^{2\pi} \int_0^{\infty} r \sin \theta (r \cos \theta + r \sin \theta) e^{-r^2} r dr d\theta \\ &= \frac{1}{4} \int_0^{\infty} \int_0^{2\pi} (\sin \theta \cos \theta + \sin^2 \theta) e^{-r^2} r^3 d\theta dr \\ &= \frac{1}{4} \int_0^{\infty} 4 \times \frac{1}{2} \times \frac{\pi}{2} e^{-r^2} r^3 dr \\ &= \frac{\pi}{8} \int_0^{\infty} t e^{-t} dt \quad (\because r^2 = t) \\ &= \frac{\pi}{8} \times 1!\end{aligned}$$

정답 : ③

【9】

$x^2 + 2xy + 5y^2 = (x \ y) \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = v^T A v$ 라 하면

$$\begin{aligned}\iint_{R^2} f(x^2 + 2xy + 5y^2) dx dy &= \frac{1}{\sqrt{\det A}} \iint_{R^2} f(u^2 + v^2) du dv \\ &= \frac{1}{2} \iint_{R^2} f(u^2 + v^2) du dv \\ &= \frac{1}{2} \int_0^{2\pi} \int_0^{\infty} f(r^2) r dr d\theta \\ &= \frac{1}{2} \int_0^{\infty} \int_0^{2\pi} f(r^2) r d\theta dr \\ &= \pi \int_0^{\infty} f(r^2) r dr = \frac{\pi}{2} \int_0^{\infty} f(t) dt \quad (\because r^2 = t) \\ &= \frac{\pi}{2} \times 4\end{aligned}$$

[다른 풀이]

$$\begin{aligned}\iint_{R^2} f(x^2 + 2xy + 5y^2) dx dy &= \iint_{R^2} f((x+y)^2 + (2y)^2) dx dy \quad (\because x+y = u, 2y = v) \\ &= \iint_{R^2} f(u^2 + v^2) \frac{1}{2} du dv \\ &= \frac{1}{2} \int_0^{2\pi} \int_0^{\infty} f(r^2) r dr d\theta \\ &= \frac{1}{2} \int_0^{\infty} \int_0^{2\pi} f(r^2) r d\theta dr \\ &= \pi \int_0^{\infty} f(r^2) r dr = \frac{\pi}{2} \int_0^{\infty} f(t) dt \quad (\because r^2 = t) \\ &= \frac{\pi}{2} \times 4\end{aligned}$$

정답 : ④

【10】

$$5x^2 - 4xy + 8y^2 = 36$$

$$\Rightarrow (x \ y) \begin{pmatrix} 5 & -2 \\ -2 & 8 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 36$$

$\Rightarrow A = \begin{pmatrix} 5 & -2 \\ -2 & 8 \end{pmatrix}$ 의 고유치는 4, 9이고 각각에 대응하는

고유벡터는 $\frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, 고유벡터는 $\frac{1}{\sqrt{5}} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ 이다.

따라서 $\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$ 라 하면 $J=1$ 이다.

$\Rightarrow 5x^2 - 4xy + 8y^2 = 36$ 을 직교대각화를 적용하면

$4u^2 + 9v^2 = 36$ 이므로 내부 영역을 R' 라 하자.

$$\Rightarrow R' : \frac{u^2}{9} + \frac{v^2}{4} \leq 1$$

$$\begin{aligned} \iint_R xy \, dA &= \iint_{R'} \frac{1}{5} (2u - v)(u + 2v) \, dudv \\ &= \frac{1}{5} \iint_{R'} 2u^2 + 3uv - 2v^2 \, dudv \\ &= \frac{1}{5} \iint_{R'} 2u^2 - 2v^2 \, dudv \quad \left(\because \iint_{R'} uv \, dudv = 0 \right) \\ &= \frac{1}{5} \iint_{R''} (18X^2 - 8Y^2) 6 \, dXdY \quad (\because u = 3X, v = 2Y) \\ &= \frac{6}{5} \iint_{R''} 10X^2 \, dXdY \\ &\quad \left(\because \iint_{X^2+Y^2 \leq 1} X^2 \, dXdY = \iint_{X^2+Y^2 \leq 1} Y^2 \, dXdY \right) \\ &= \frac{6}{5} \times 10 \times \frac{1}{2} \iint_{R''} (X^2 + Y^2) \, dXdY \\ &= 6 \int_0^{2\pi} \int_0^1 r^3 \, dr d\theta \\ &= 6 \times 2\pi \times \frac{1}{4} = 3\pi \end{aligned}$$

[다른 풀이]

$$R : 5x^2 - 4xy + 8y^2 \leq 36$$

$$\Rightarrow \left(\frac{3}{\sqrt{2}}x \right)^2 + \left(\frac{1}{\sqrt{2}}x - 2\sqrt{2}y \right)^2 \leq 36 \text{이므로}$$

$$u = \frac{3}{\sqrt{2}}x, \quad v = \frac{1}{\sqrt{2}}x - 2\sqrt{2}y \text{라 하면}$$

$$x = \frac{\sqrt{2}}{3}u, \quad y = \frac{1}{2\sqrt{2}} \left(\frac{1}{3}u - v \right) \text{이고}$$

$$J = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} \frac{\sqrt{2}}{3} & 0 \\ \frac{1}{6\sqrt{2}} & -\frac{1}{2\sqrt{2}} \end{vmatrix} = -\frac{1}{6} \text{이다.}$$

$$\Rightarrow R' : u^2 + v^2 \leq 36$$

$$\begin{aligned} \iint_R xy \, dA &= \iint_{R'} \frac{\sqrt{2}}{3}u \times \frac{1}{2\sqrt{2}} \left(\frac{1}{3}u - v \right) \times \frac{1}{6} \, dudv \\ &= \frac{1}{36} \iint_{R'} \left(\frac{1}{3}u^2 - uv \right) \, dudv \\ &= \frac{1}{36} \iint_{R'} \frac{1}{3}u^2 \, dudv \quad \left(\because \iint_{R'} uv \, dudv = 0 \right) \\ &= \frac{1}{36} \times \frac{1}{3} \times \frac{1}{2} \iint_{R'} u^2 + v^2 \, dudv \end{aligned}$$

$$\begin{aligned} &= \frac{1}{6^3} \int_0^{2\pi} \int_0^6 r^3 \, dr d\theta \\ &= \frac{1}{6^3} \times \frac{6^4}{4} \times 2\pi = 3\pi \end{aligned}$$

정답 : ③